Jump Modelling with Dirichlet Process Mixture

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Motivation

Jump Diffusion Models

Jump diffusion models are introduced to:

- Model the sudden and extreme changes (Merton, 1976; Bates, 1996; Kou, 2002, 2007),
- Capture the fat tails of return distribution,
- Adding jumps to stochastic volatility (SV) models better explain the empirical implied volatility (IV) smile (Pan, 2002; Eraker *et al.*, 2003; Eraker, 2004).
- Hence provide better specification for option pricing.

Classic SVJ model:

Let p_t be the logarithm of the asset price P_t at time t and V_t be the latent variance.

$$\begin{aligned} dp_t &= \mu_t dt + (V_t)^{1/2} dW_t^p + Z_t dN_t, \\ d(\log(V_t)) &= \kappa(\theta - \log(V_t))dt + \gamma dW_t^v, \end{aligned} \tag{1}$$

where dN_t represents the jump occurrence and Z_t is the jump size, which is assumed to follow a particular distribution.

Some further investigation of jump structure: dynamic jumps (Maheu and McCurdy, 2004); jump clustering (Aït-Sahalia *et al.*, 2015); jump directions (Maneesoonthorn *et al.*, 2017).

Motivation: Simulation Illustration

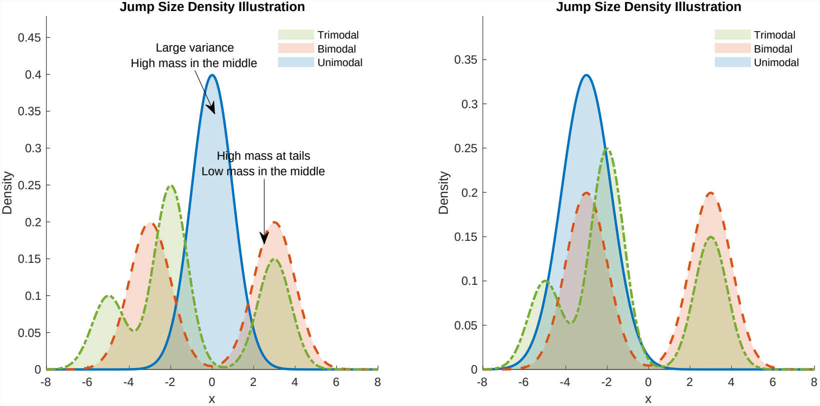

Figure 1: Simulation illustration for the motivation of proposing mixed jumps. The red dashed line shows the bimodal distribution, the green dot-dashed line shows a trimodal distribution and the blue solid line is a single normal density.

Motivation

The disadvantage of parametric SVJ model:

- Assuming a single particular distribution for jump size - lack of flexibility;
- Jump size distribution could have large variance and relatively high mass around "small value";
- The misspecification in jump distribution will be reflected in less accurate option prices and IV.

Contribution

We propose a new approach of modelling jump size density using Bayesian nonparametric technique - Dirichlet process mixture (DPM).

We show that our proposed model can

- Model the jump size distribution flexibly;
- Recover the empirical jump size distribution.

Model Specification

Discrete time static SV-DPM jump model

On day t, let r_t be the log return and ,h_t be the log-variance of return and ΔN_t represents the jump arrival and Z_t is the jump size.

$$\begin{aligned}
r_t &= \exp(h_t/2)\varepsilon_t^r + \Delta N_t Z_t, \\
h_t &= \phi_0 + \phi_1 h_{t-1} + \sigma_t^h \varepsilon_t^h, \\
\Delta N_t &\sim \text{Bernoulli}(\delta), \\
Z_t &\sim \sum_{k=1}^{\infty} w_k N(\mu_k, \sigma_k^2), \\
w_k &= v_k \prod_{l=1}^{k-1}(1 - v_l), \text{ with } v_k \sim \text{Beta}(1, \alpha),
\end{aligned} \tag{2}$$

where $\varepsilon_t^r, \varepsilon_t^{BV}, \varepsilon_t^h$ follow $N(0,1)$. We model jump size Z_t distribution as a stick-breaking process (SBP) (Sethuraman, 1994).

Further development: Practical motivation

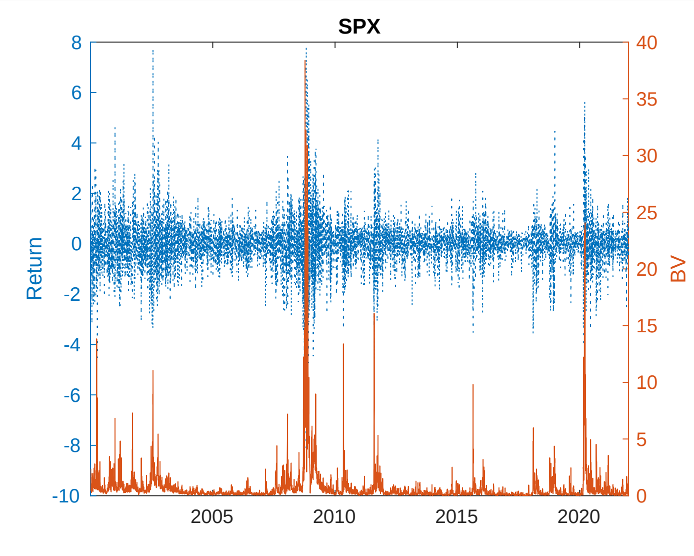

Figure 2: Empirical return and bipower variation for SPX.

Discrete time dynamic SV-DPM jump model

Incorporating bipower variation:

$$BV_t = \frac{\pi}{2}\left(\frac{N}{N-1}\right)\sum_{i=2}^{N}|r_{t_i}||r_{t_{i+1}}| \xrightarrow{p} IV_t, \text{ as } N \to \infty, \tag{3}$$

where $r_{t_i} = p_{t_{i+1}} - p_{t_i}$ is the ith of N observed returns between trading day t and $t+1$ (Barndorff-Nielsen and Shephard, 2004).

The specification of jump size becomes:

$$\begin{aligned} Z_t &\sim \sum_{k=1}^{\infty} w_k N(\mu_k(\bar{BV_t})^{1/2}, \sigma_k^2), \\ w_k &= v_k \prod_{l=1}^{k-1}(1-v_l), \text{ with } v_k \sim \text{Beta}(1,\alpha), \\ (\bar{BV_t})^{1/2} &= \left[\frac{1}{n}\sum_{m=1}^{n}(BV_{t-m})\right]^{1/2} \end{aligned} \tag{4}$$

Leverage Effect

Inference

Joint Posterior

Let $\Theta = \{\phi_0, \phi_1, \sigma_h^2, \sigma_{BV}^2\}$ and $\boldsymbol{\Phi} = \{\boldsymbol{w}, \mu, \boldsymbol{\sigma^2}, \delta\}$ where $\mu = \{\mu_1, \dots, \mu_\infty\}$ and $\boldsymbol{\sigma^2} = \{\sigma_1^2, \dots, \sigma_\infty^2\}$.

For dynamic DPM model:

$$
\begin{aligned}
P(\boldsymbol{\Theta}, \boldsymbol{\Phi}, \boldsymbol{h}, \boldsymbol{\Delta N}, \boldsymbol{Z} | \boldsymbol{r}, \boldsymbol{(B\bar{V})}^{1/2}) \propto\ & P(\phi_0, \phi_1 | \sigma_h^2) P(\sigma_h^2) P(\delta) P(h_0) \\
& \prod_{t=1}^{T} \Big[P(\Delta N_t | \delta) P(h_t | h_{t-1}, \phi_0, \phi_1, \sigma_h^2) P(r_t | h_t, Z_t, \Delta N_t) \\
& \prod_{k=1}^{\infty} P(\mu_k, \sigma_k^2) P(w_k) P(Z_t | \mu_k (B\bar{V}_t)^{1/2}, \sigma_k^2) \Big]
\end{aligned}
\tag{5}
$$

Slice Sampler

An auxiliary variable u_t is introduced who is uniformly distributed conditional on $\boldsymbol{w}$ and indicator variable of jump mixture S_t^z.

$$u_t | S_t^z, \boldsymbol{w} \sim U(0, w_{S_t^z}). \tag{6}$$

The joint distribution of u_t and Z_t is:

$$\begin{aligned} p(u_t, Z_t | \cdot) &= \sum_{k=1}^{\infty} w_k U(u_t | 0, w_k) N(Z_t | \mu_k, \sigma_k^2), \\ &= \sum_{k=1}^{\infty} I(u_t < w_k) N(Z_t | \mu_k, \sigma_k^2), \\ &= \sum_{k \in A_t} N(Z_t | \mu_k, \sigma_k^2) \end{aligned} \tag{7}$$

where $A_t = \{k : u_t < w_k\}$. Notably, A_t here is a set with finite elements (Neal, 2003; Walker, 2007). DAGs

MCMC Algorithm

With total M iterations, the MCMC algorithm is:

Algorithm 1 Full Gibbs sampling algorithm

Require: prior, $\boldsymbol{r}$

Initialize $(i=0)$ $\boldsymbol{\Theta}, \boldsymbol{S}$ for initialization of $\boldsymbol{h}$;

Initialize $(i=0)$ $(\mu_{\boldsymbol{S}_t^z}, \sigma^2_{\boldsymbol{S}_t^z}), \boldsymbol{S}^z$ for initialization of $\boldsymbol{Z}$.

Initialize $(i=0)$ δ

For each MCMC iteration i

1. Sample $\boldsymbol{\Delta N}^{(i)} | \boldsymbol{h}^{(i-1)}, \boldsymbol{r}, \boldsymbol{Z}^{(i-1)}, \delta^{(i-1)}$
2. Sample $\delta^{(i)} | \boldsymbol{\Delta N}^{(i)}$
3. Sample $\boldsymbol{w}^{(i)} | \boldsymbol{S}_z^{(i-1)}, \boldsymbol{u}^{(i)} | \boldsymbol{w}^{(i)}, \boldsymbol{S}_z^{(i-1)}$
4. Sample $(\mu_{\boldsymbol{S}_t^z}, \sigma^2_{\boldsymbol{S}_t^z})^{(i)} | \boldsymbol{S}_z^{(i-1)}, \boldsymbol{Z}^{(i-1)}$
5. Sample $\boldsymbol{S}_z^{(i)} | \boldsymbol{Z}^{(i-1)}, (\mu_{\boldsymbol{S}_t^z}, \sigma^2_{\boldsymbol{S}_t^z})^{(i)}, \boldsymbol{w}^{(i)}, \boldsymbol{u}^{(i)}$
6. Sample $\boldsymbol{Z}^{(i)} | \boldsymbol{h}^{(i-1)}, \boldsymbol{r}, \boldsymbol{\Delta N}^{(i)}, \boldsymbol{S}_z^{(i)}, (\mu_{\boldsymbol{S}_t^z}, \sigma^2_{\boldsymbol{S}_t^z})^{(i)}$
7. Sample $\boldsymbol{h}^{(i)} | \boldsymbol{r}, \boldsymbol{\Delta N}^{(i)}, \boldsymbol{Z}^{(i)}, \boldsymbol{S}^{(i-1)}, \boldsymbol{\Theta}^{(i-1)}$ ▷ using 7-component mixture (Kim *et al.*, 1998).
8. Sample $\boldsymbol{\Theta}^{(i)} | \boldsymbol{h}^{(i)}$
9. Sample $\boldsymbol{S}^{(i)} | \boldsymbol{h}^{(i)}, \boldsymbol{r}$ ▷ $\boldsymbol{S}$ is the indicator for seven point mixture.
10. Go to step (1) and repeat for M iterations.

return $\boldsymbol{h}, \boldsymbol{Z}, \boldsymbol{\Delta N}, \boldsymbol{\Theta}$

Simulation: Model Sensitivity

Simulation Analysis

The purpose of this simulation is to investigate the model performance in d**etecting true jumps** under different true DGPs:

- when true jump intensity changes;
- when true jump size changes;
- how our model performs compared with parametric SVJ model where jump size $Z_t \sim N(\mu_z, \sigma_z^2)$.

True Data Generating Process

We generate $T = 5000$ observations from the following true DGP:

$$\begin{aligned}
r_t &= \exp(h_t/2)\varepsilon_t^r + \Delta N_t Z_t, \\
h_t &= -0.06 + 0.94 h_{t-1} + 0.38\varepsilon_t^h, \\
\log(BV_t) &= h_t + 0.4\varepsilon_t^{BV}, \\
B\bar{V}_t &= \frac{1}{5}\sum_{m=1}^{5} BV_{t-m}, \\
\Delta N_t &\sim \text{Bernoulli}(\delta), \\
Z_t &\sim 0.6 \times N(-\mu_z e^{h_t/2}, 0.8) + 0.4 \times N(\mu_z e^{h_t/2}, 0.6).
\end{aligned} \tag{8}$$

We compare the model performance on a various range of $\delta = \{1\%, 5\%, 10\% 15\%\}$ and $\mu_z = \{2,\ 2.5,\ 3,\ 3.5\}$ away from the conditional mean of return distribution.

Simulation Results: AUC

Area under the ROC curve (AUC)			
	Dynamic DPM	Static DPM	Parametric SVJ
A: Jump Intensity Analysis ($\mu_z = 3$)			
$\delta = 1\%$	0.8960	0.8791	**0.9043**
$\delta = 5\%$	**0.9054**	0.8883	0.8819
$\delta = 10\%$	**0.9004**	0.8816	0.8788
$\delta = 15\%$	**0.9208**	0.9018	0.9018
B: Jump Size Analysis ($\delta = 10\%$)			
$\mu_z = 2$	0.7966	0.7932	**0.7974**
$\mu_z = 2.5$	**0.8491**	0.8270	0.8287
$\mu_z = 3$	**0.9004**	0.8816	0.8788
$\mu_z = 3.5$	**0.9363**	0.9160	0.9171

Table 1: Area under the ROC curve (AUC) for both jump intensity analysis and jump size analysis under simulation. The larger AUC, the more accurate the model is in terms of detecting true jumps. The bold numbers are the maximum of each cases.

Simulation Summary

- For the true jump intensity $\delta = 5\%, 10\%, 15\%$, DPM models are more accurate in detecting true jumps, but it suffers from small sample problem ($\delta = 1\%$) and is slightly worse than parametric SVJ model;
- If the true jump size $\mu_z \geq 2.5$, dynamic DPM model is best in detecting true jumps, but if the true jump size $\mu_z = 2$, which is not considered as in the extreme tails, parametric SVJ model performs better.

Empirical Analysis

Data

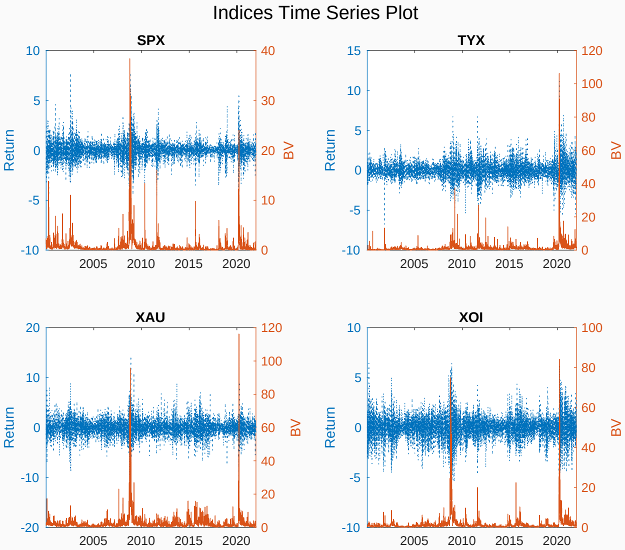

Figure 3: Time series plots of four financial indices. SPX - S&P 500 index, TYX - Treasury yield index (30 year bond), XAU - gold and silver index and XOI - oil index.

Estimated jump size distribution

Estimated jump size distributions for financial indices		
Parameters	Posterior Mean/Mode (for K)	90% Posterior Interval/Range (for K)
	SPX	
K	2	[2,10]
μ_z	1.2220; -1.5656	(0.9236, 1.5885); (-1.9709, -1.1900)
σ_z	0.4022; 0.4787	(0.3192,0.5044); (0.3703, 0.5934)
	TYX	
K	2	[2,10]
μ_z	-1.2943; 1.5057	(-1.8216, -0.8522); (1.2094, 1.8179)
σ_z	0.5380; 0.4849	(0.3946,0.7217); (0.3668, 0.6322)
	XAU	
K	2	[2,10]
μ_z	2.0420; -1.6369	(1.6303. 2.4210); (-1.9333,-1.3232)
σ_z	0.5586; 0.6294	(0.4038, 0.7638); (0.4344, 0.8770)
	XOI	
K	2	[2,6]
μ_z	-1.7014; 1.1320	(-2.0487, -1.3210); (0.8083, 1.4869)
σ_z	0.5556; 0.5728	(0.4019, 0.7758); (0.4238,0.7506)

Table 2: Estimated jump component K, and the corresponding mean and variance of jump size μ_z, σ_z. We report the posterior mean/mode (for K) and 90% posterior interval or range for K.

Results I: Estimated jump intensity

Estimated Jump Intensity for Financial Indices				
	SPX	TYX	XAU	XOI
Parametric	0.0975	0.1482	0.1900	0.1329
	(0.0661, 0.1310)	(0.1054, 0.1965)	(0.1533 ,0.2345)	(0.0955, 0.1756)
DPM	0.1071	0.1107	0.1341	0.1249
	(0.0757, 0.1435)	(0.0744, 0.1538)	(0.1031, 0.1653)	(0.0876, 0.1645)

Table 3: Post-convergence **posterior mean of jump intensity** δ and their 90% posterior interval.

Model Implied Jump Density: SPX

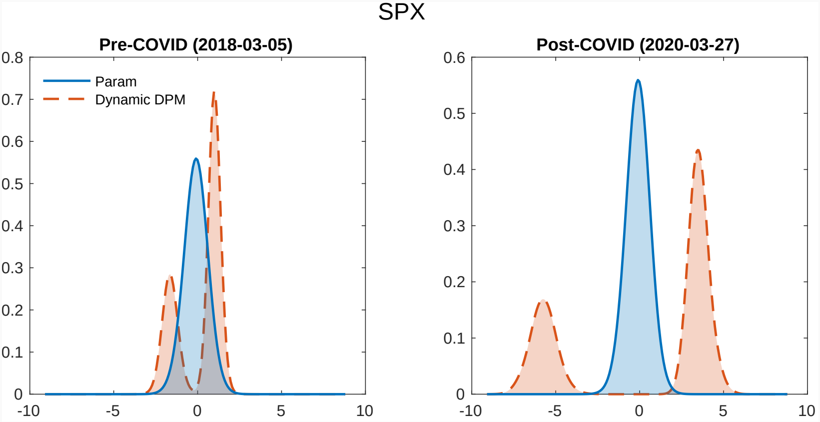

Figure 4: The **model implied jump density** of both parametric and DPM jump models for SPX. The blue solid line represents the jump kernel estimated from **parametric** jump model. The red dashed line represents the estimated jump kernel from **dynamic DPM** model.

Model Implied Jump Density: XAU

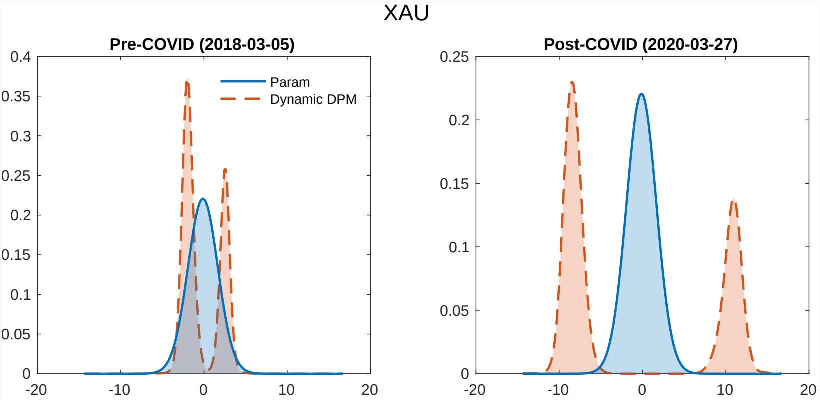

Figure 5: The **model implied jump density** of both parametric and DPM jump models for XAU. The blue solid line represents the jump kernel estimated from **parametric** jump model. The red dashed line represents the estimated jump kernel from **dynamic DPM** model.

Predictive Density of Returns

Let information at time t be $\mathscr{F}_t$, for total M MCMC iterations, we obtain **one-step** ahead predictive density of return via expanding window:

$$\hat{P}(r_{t+1}|\mathscr{F}_t) = \frac{1}{M}\sum_{i=1}^{M}\hat{P}(r_{t+1}|\mathscr{F}_t,(\bar{BV}_{t+1}^{(i)})^{1/2})^{(i)}, \tag{9}$$

where

$$\hat{P}(r_{t+1}|\mathscr{F}_t,(\bar{BV}_{t+1}^{(i)})^{1/2})^{(i)} \sim (1-\delta^{(i)})N(0,e^{h_{t+1}^{(i)}})+ \\ \delta^{(i)}\sum_{k=1}^{K^{(i)}} w_k^{(i)} N\Big(\mu_k^{(i)}(\bar{BV}_{t+1}^{(i)})^{1/2}, e^{h_{t+1}^{(i)}}+(\sigma_k^2)^{(i)}\Big). \tag{10}$$

Predictive Density Evaluation

We evaluate the predictive density using strictly proper scoring rules (Gneiting and Raftery, 2007; Diks *et al.*, 2011; Gneiting and Ranjan, 2011).

Log score (LS):

$$S_{LS}(P_{t+1}, r_{t+1}) = \log(p(r_{t+1}|\mathscr{F}_t)) \tag{11}$$

Censored likelihood score (CLS):

$$\begin{aligned} S_{CLS}(P_{t+1}, r_{t+1}) = {} & I(r_{t+1} \in A)\log(p(r_{t+1}|\mathscr{F}_t)) \\ & + I(r_{t+1} \in A^c)\log\Big(\int_{r \in A^c} p(r|\mathscr{F}_t)dr\Big), \end{aligned} \tag{12}$$

where A is the interested region.

Prediction Evaluation

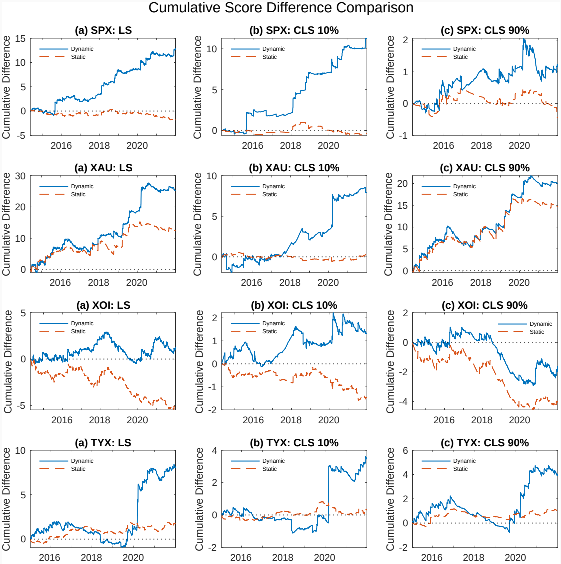

Figure 6: Score evaluation of predictive density for all indices.

Empirical Application in Implied Volatility Smile

Verify our model using implied volatility (IV) smile:

1. Under the no-arbitrage conditions, assume the r_t and h_t following the dynamic SV-DPM jump model and parametric SVJ model;
2. Simulate European call options and their corresponding IV with the estimated parameters of the models;
3. As the average jump size can be proxied by the slope of IV smile, we normalize IV with its own at-the-money options and compare with the empirical IV curve (Cont, 2010; Yan, 2011).

Definition

Empirical Application in Implied Volatility Smile

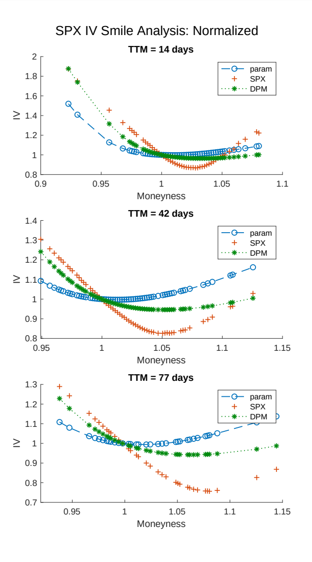

Figure 7: Normalized IV comparison for SPX index.

Summary

The model we proposed here provides:

- Dynamic DPM provides better estimation results than parametric SVJ and static DPM;
- Recover the empirical jump size distribution driven by data;
- Provides better predictive return density than parametric SVJ model, especially during volatile periods.
- The extension with leverage effects can be found in the paper

Questions?

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Appendix

Implied Volatility: Definition

Implied volatility is the volatility that is implied by the market option price. Let c_t^m be market call option price and solve for $\sigma = \hat{\sigma}$ such that

$$c_t^m = c_t^{BS}(\sigma), \tag{13}$$

where c_t^{BS} is the call option price calculated using Black-Scholes model (Black and Scholes, 1973). Application

Models

Assume the stock price follows the structure as below:

1. SV model
2. Parametric SVJ model
3. SV model with mixture jumps

Let X be strike price, r_f be risk-free rate and T_m be time-to-maturity. The call option price can be simulated via Monte Carlo simulation as:

$$C(S_T, X, T_m) = \frac{1}{(1+r_f)^{T_m}} E^Q[\max(S_{T_m} - X, 0)]. \tag{14}$$

DAGs with Slice Sampler

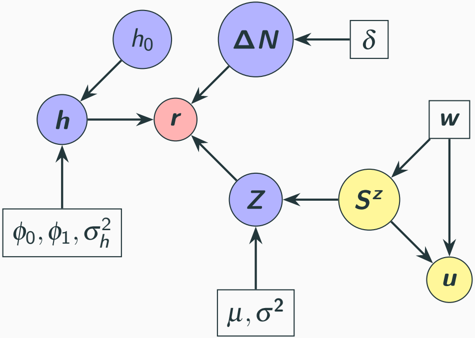

Figure 8: DAGs of the DPM jump model structure. Red circles represents observed data, purple circles are latent variables. Parameters are represented as rectangular. Yellow circles represents the auxiliary variables where $\boldsymbol{u}$ is introduced for slice sampler and $\boldsymbol{S^z}$ is an indicator functions for components.

slice sampler

Leverage Effect

"Leverage effect": the asset's volatility is negatively correlated with the asset's returns (Black, 1976; Christie, 1982).

$$
\begin{aligned}
r_t &= \exp(h_t/2)\varepsilon_t^r + \Delta N_t Z_t, \\
h_t &= \phi_0 + \phi_1 h_{t-1} + \sigma_t^h \varepsilon_t^h, \\
\begin{pmatrix} \varepsilon_t^r \\ \varepsilon_t^h \end{pmatrix} &\sim N\left(0, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}\right) \\
\Delta N_t &\sim \text{Bernoulli}(\delta), \\
Z_t &\sim \sum_{k=1}^{\infty} w_k N(\mu_k (\bar{BV}_t)^{1/2}, \sigma_k^2).
\end{aligned}
\tag{15}
$$

Predictive Density Evaluation with leverage effect

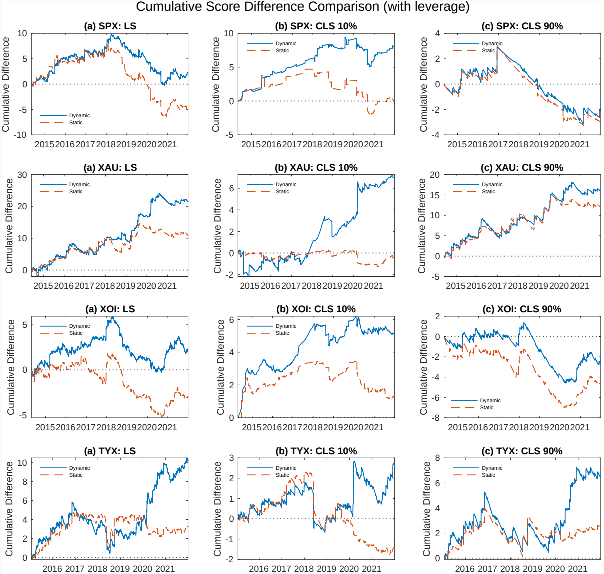

Figure 9: Score evaluation of predictive density for all indices with leverage effects.

Simulation Prior

	Simulation Analysis		
	Prior Spec.	Mean	Variance
SV model parameters			
$(\phi_0, \phi_1)'$	$\text{N}(\boldsymbol{m_h = (0, 0.9)'}, \boldsymbol{H_h^{-1} = (1000, 1000)'})$	(0,0.9)'	(1000,1000)'
$\boldsymbol{\sigma_h^{-2}}$	$\text{Gamma}(\frac{a_h}{2} = 2.01, \frac{b_h}{2} = 2.01)$	1	0.99
Jump model parameters			
δ	$\text{Beta}(\alpha_0 = 2, \beta_0 = 5)$	0.2857	0.0255
μ	$\text{N}(m = 0, h^{-1} = 1000)$	0	1000
$\boldsymbol{\sigma^{-2}}$	$\text{Gamma}(\frac{a}{2} = 2.01, \frac{b}{2} = 2.01)$	1	0.99
$\boldsymbol{w}$	$\text{SBP}(\alpha = 1)$	-	-

Table 4: Prior specifications for parameters in simulation analysis

Empirical Analysis Prior

Prior distribution of static parameters		
SV model parameters:		
(ϕ_0,ϕ_1)	$\mathrm{N}(\boldsymbol{m_h},\boldsymbol{H_h}^{-1})$	$\{\boldsymbol{m_h}=[0,0.9],\boldsymbol{H_h}=[0.001,0.001]\}$
σ_h^2	$\mathrm{IG}(a_h/2,b_h/2)$	$\{a_h=5.01,b_h=1.01\}$
Jump component parameters:		
δ	$\mathrm{Beta}(\alpha_0,\beta_0)$	$\{\alpha_0=10,\beta_0=130\}$
$\boldsymbol{w}$	$\mathrm{SBP}(\alpha)$	$\alpha=1$
Parametric SVJ:		
μ	$\mathrm{N}(m,h^{-1})$	$\{m=-3,h=1\}$
σ^2	IG(a/2,b/2)	$\{a=b=2.01\}$
DPM Bimodal prior:		
μ	$p_\mu N(m_1,h^{-1})+(1-p_\mu)N(m_2,h^{-1})$	$\{m_1=-3,m_2=3,h=1,p_\mu=0.5\}$
σ^2	IG(a/2,b/2)	$\{a=10.01,b=3.01\}$

Table 5: Prior for empirical analysis